

OpenStax Calculus Volume 2

Section 2.6 Exercises: Moments and Centers of Mass

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For the following exercises, calculate the center of mass for the collection of masses given.

1. $m_1 = 2$ at $x_1 = 1$ and $m_2 = 4$ at $x_2 = 2$
2. $m_1 = 1$ at $x_1 = -1$ and $m_2 = 3$ at $x_2 = 2$
3. $m = 3$ at $x = 0, 1, 2, 6$
4. Unit masses at $(x, y) = (1, 0), (0, 1), (1, 1)$
5. $m_1 = 1$ at $(1, 0)$ and $m_2 = 4$ at $(0, 1)$
6. $m_1 = 1$ at $(1, 0)$ and $m_2 = 3$ at $(2, 2)$

For the following exercises, compute the center of mass $\bar{x}$.

7. $\rho = 1$ for $x \in (-1, 3)$
8. $\rho = x^2$ for $x \in (0, L)$
9. $\rho = 1$ for $x \in (0, 1)$ and $\rho = 2$ for $x \in (1, 2)$
10. $\rho = \sin x$ for $x \in (0, \pi)$
11. $\rho = \cos x$ for $x \in (0, \frac{\pi}{2})$
12. $\rho = e^x$ for $x \in (0, 2)$
13. $\rho = x^3 + xe^{-x}$ for $x \in (0, 1)$
14. $\rho = x \sin x$ for $x \in (0, \pi)$
15. $\rho = \sqrt{x}$ for $x \in (1, 4)$
16. $\rho = \ln x$ for $x \in (1, e)$

For the following exercises, compute the center of mass $(\bar{x}, \bar{y})$. Use symmetry to help locate the center of mass whenever possible.

17. $\rho = 7$ in the square $0 \leq x \leq 1, 0 \leq y \leq 1$
18. $\rho = 3$ in the triangle with vertices $(0, 0)$, $(a, 0)$, and $(0, b)$

19. $\rho = 2$ for the region bounded by $y = \cos(x)$, $y = -\cos(x)$, $x = -\frac{\pi}{2}$, and $x = \frac{\pi}{2}$

For the following exercises, use a calculator to draw the region, then compute the center of mass $(\bar{x}, \bar{y})$. Use symmetry to help locate the center of mass whenever possible.

20. [T] The region bounded by $y = \cos(2x)$, $x = -\frac{\pi}{4}$, and $x = \frac{\pi}{4}$
21. [T] The region between $y = 2x^2$, $y = 0$, $x = 0$, and $x = 1$
22. [T] The region between $y = \frac{5}{4}x^2$ and $y = 5$
23. [T] Region between $y = \sqrt{x}$, $y = \ln(x)$, $x = 1$, and $x = 4$
24. [T] The region bounded by $y = 0$, $\frac{x^2}{4} + \frac{y^2}{9} = 1$
25. [T] The region bounded by $y = 0$, $x = 0$, and $\frac{x^2}{4} + \frac{y^2}{9} = 1$
26. [T] The region bounded by $y = x^2$ and $y = x^4$ in the first quadrant

For the following exercises, use the theorem of Pappus to determine the volume of the shape.

27. Rotating $y = mx$ around the x -axis between $x = 0$ and $x = 1$
28. Rotating $y = mx$ around the y -axis between $x = 0$ and $x = 1$
29. A general cone created by rotating a triangle with vertices $(0, 0)$, $(a, 0)$, and $(0, b)$ around the y -axis. Does your answer agree with the volume of a cone?
30. A general cylinder created by rotating a rectangle with vertices $(0, 0)$, $(a, 0)$, $(0, b)$, and (a, b) around the y -axis. Does your answer agree with the volume of a cylinder?
31. A sphere created by rotating a semicircle with radius a around the y -axis. Does your answer agree with the volume of a sphere?

For the following exercises, use a calculator to draw the region enclosed by the curve. Find the area M and the centroid $(\bar{x}, \bar{y})$ for the given shapes. Use symmetry to help locate the center of mass whenever possible.

32. [T] Quarter-circle: $y = \sqrt{1 - x^2}$, $y = 0$, and $x = 0$
33. [T] Triangle: $y = x$, $y = 2 - x$, and $y = 0$
34. [T] Lens: $y = x^2$ and $y = x$
35. [T] Ring: $y^2 + x^2 = 1$ and $y^2 + x^2 = 4$
36. [T] Half-ring: $y^2 + x^2 = 1$, $y^2 + x^2 = 4$, and $y = 0$
37. Find the generalized center of mass in the sliver between $y = x^a$ and $y = x^b$ with $a > b$. Then, use the Pappus theorem to find the volume of the solid generated when revolving around the y -axis.
38. Find the generalized center of mass between $y = a^2 - x^2$, $x = 0$, and $y = 0$. Then, use the Pappus theorem to find the volume of the solid generated when revolving around the y -axis.
39. Find the generalized center of mass between $y = b \sin(ax)$, $x = 0$, and $x = \frac{\pi}{a}$. Then, use the Pappus theorem to find the volume of the solid generated when revolving around the y -axis.

40. Use the theorem of Pappus to find the volume of a torus (pictured here). Assume that a disk of radius a is positioned with the left end of the circle at $x = b$, $b > 0$, and is rotated around the y-axis.

[Figure: This figure is a torus. It has inner radius of b . Inside of the torus is a cross section that is a circle. The circle has radius a .]

41. Find the center of mass $(\bar{x}, \bar{y})$ for a thin wire along the semicircle $y = \sqrt{1 - x^2}$ with unit mass. (Hint: Use the theorem of Pappus.)